

Venkat Vaddula

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WORK EXPERIENCE

Senior Data Engineer, Climate LLC, Seattle, WA

Feb 2023 – Present

- Refactored an NLP matching algorithm into a multiple ML model by developing fuzzy metric selection logic and productionized the targeted model that increased the valid response rate of custom user input text by 15%.
- Optimized a heuristic algorithm that validates business rules for customer programs and improved performance by 200% maintaining accuracy to enable productionization using AWS ECS.
- Deployed at-scale text processing models as internal conda and pip packages using Airflow that are run daily on docker images to create hive data tables in data warehouse and then collaborated with downstream science research as well as product and engineering teams to build dashboards on Looker / Tableau.
- Streamlined science model deployment by writing unit and E2E tests in collaboration with data and atmospheric scientists also developed and supported GitLab CI/CD pipeline templates that enabled automated and rapid development and testing of ML models.
- Improved several operational processes like developing custom environments that resulted in 2x faster data science workspace startups, optimizing spark configs and partitioning datasets by use case and size, setting up utils packages to reduce redundancies and improve reusability.

Senior Data Analyst, Climate LLC, Chicago, IL

Oct 2020 – Feb 2023

- Developed a contextual clustering model utilizing geospatial and temporal features that increased the amount of clean model training data by more than 25%.
- Technical lead for text matching and novel geospatial data pipeline program spaces responsible for developing goals, project roadmaps and leading milestone stakeholder discussions ensuring model and data delivery to support making business decisions and validating customer data.
- Collaborated with cross functional teams to develop organization level coding, documentation, and packaging standards and best practices that streamlined operations and enabled faster reviews and model deployment.

Data Analyst, Climate LLC, St. Louis, MO

Jun 2019 – Sep 2020

- Identified and developed a POC fix to a top data quality issue in the production pipeline, that measurably increased accuracy of cutting edge predictive model by 7%.
- Diagnosed several root causes impacting production data and implemented enhancements by collaborating with data engineers and scientists performing sensitivity analysis of predictive analytics and presented results at multiple venues from weekly technical research discussions to organization level data summits.

Data Analyst, Rose International Inc, St. Louis, MO (contractor for Climate LLC)

Sep 2018 – May 2019

- Improved a product name matching algorithm using various text processing regex and NLP techniques that resulted in a 10% decrease in unusable data for R&D. Also modularized and refactored codebase into a conda package to streamline usage and future improvements.
- Built a Decision Tree model that detected the classes of typos users make in product names with an accuracy of 87%, Deployed model as a looker dashboard to enable product leadership in prioritizing goal and projects.

TECHNICAL SKILLS

Programming Languages: Python (6+ yrs), SQL (6+ yrs), Java (4+ yrs), JavaScript (4+ yrs)

Frameworks/Libraries/Databases: Spark/PySpark, Pandas, Geopandas, NLP, Hive, Scikit-learn, Pandas, Matplotlib, Postgres, Vetrica, AWS (S3, Aurora, ECS), GCP BigQuery, Airflow, Git

EDUCATION

Masters in Computer Science (specialization in Data Science)

The University of Kansas, Kansas

RECOGNITION

- “Better Because of You” Peer Recognition Award (Climate 2019, 2020, 2021, 2022 & 2023)
- Gold Pinnacle Award - Performance that drives exceptional impact and extraordinary results (Climate 2022)